

A. Gaussians and Hessians

PSO
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② $f(x) = \frac{1}{2} x^T A x + b^T x$, A Symmetric, $b \in \mathbb{R}^n$

For the properties of gradients: $\nabla_x (f(x) + g(x)) = \nabla_x f(x) + \nabla_x g(x)$

Therefore: $\nabla_x f(x) = \nabla_x \left[\frac{1}{2} x^T A x + b^T x \right] = \nabla_x \left[\frac{1}{2} x^T A x \right] + \nabla_x [b^T x]$

It was already demonstrated that: $\nabla_x \left[\frac{1}{2} x^T A x \right] = A x$ if A is symmetric (which is the case)

$\nabla_x [b^T x] = b$

Therefore: $\nabla_x f(x) = \frac{1}{2} \nabla_x [x^T A x] + \nabla_x [b^T x] = \frac{1}{2} \cdot 2 A x + b = A x + b$

③ For the composition of derivatives (and the chain rule), for a generic x_i , component/entry of x

~~$\nabla_x f(x) = \nabla_x (g(h(x)))$~~ $= \frac{\partial f(x)}{\partial x_i} = \frac{\partial g(h(x))}{\partial x_i} = \frac{\partial g(h(x))}{\partial h(x)} \cdot \frac{\partial h(x)}{\partial x_i}$

Therefore: $\nabla_x f(x) = \left[\frac{\partial g(h(x))}{\partial h(x)} \cdot \frac{\partial h(x)}{\partial x_i} \right]_{i=1 \dots n}$ for i from 1 to n , ~~keeping the length of the vector~~
assuming $x \in \mathbb{R}^n$

④ From point ②, $\nabla_x f(x) = A x + b$

Therefore: $\nabla_x f(x) = \begin{bmatrix} -2x_1^T \\ -2x_2^T \\ \vdots \\ -2x_n^T \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} + b = \begin{bmatrix} 2x_1^T x + b_1 \\ 2x_2^T x + b_2 \\ \vdots \\ 2x_n^T x + b_n \end{bmatrix}$

It has already been shown that: $\nabla_x^2 f(x) = \left[\nabla_x (\nabla_x f(x))_1 \quad \nabla_x (\nabla_x f(x))_2 \quad \dots \quad \nabla_x (\nabla_x f(x))_n \right]$

It follows:

$\nabla_x^2 f(x) = \left[\nabla_x (2x_1^T x + b_1) \quad \nabla_x (2x_2^T x + b_2) \quad \dots \quad \nabla_x (2x_n^T x + b_n) \right]$
 $= \begin{bmatrix} 2x_1 & 2x_2 & \dots & 2x_n \end{bmatrix} = A$ (since A is symmetric) $\Rightarrow \nabla_x^2 f(x) = A$

⑤ Function $f(x)$ can be written as $g(h(x))$ with $h(x) = x^T x$ and $h: \mathbb{R}^n \rightarrow \mathbb{R}$
It follows that the gradient of the function can be computed as in point ③

~~$\nabla_x f(x) = g'(h(x)) \nabla h(x)$~~ $= \underline{g'(x^T x)} \cdot \underline{2}$, since $\nabla_x h(x) = \nabla_x (x^T x) = 2x$

Similar to the previous case: ~~vector, cannot be squared~~

$\nabla_x^2 f(x) = \left[\nabla_x (g'(x^T x))_1 \quad \nabla_x (g'(x^T x))_2 \quad \dots \quad \nabla_x (g'(x^T x))_n \right]$

In which each term is: $\nabla_x^2 g(x^T x)$ ~~section: take a look at each element~~

Therefore $\nabla_x^2 f(x) = \nabla_x^2 g(x^T x)$

~~not stay at the end~~
becomes $x^T x^T$

$\frac{\partial^2 g(x^T x)}{\partial x_i \partial x_j} = \frac{\partial^2 g(h(x))}{\partial (h(x))^2} \cdot \frac{\partial h(x)}{\partial x_i} \cdot \frac{\partial h(x)}{\partial x_j}$
 $\downarrow \quad \downarrow \quad \downarrow$
 $g''(x^T x) \quad 2x_i \quad 2x_j$
 $\nabla_x^2 f(x) = g''(x^T x) \cdot 2 \cdot 2x$